

Granuloma Annulare

The typical histopathologic appearance of granuloma annulare is characterized by single or multiple foci of inflammation featuring a central area of altered collagen, known as necrobiosis. This necrobiotic core is surrounded by a rim of palisaded histiocytes (Fig. 43-4). The necrotic centers are typically oval, slightly basophilic, and lack cellular nuclei. They show disruption and loss of normal collagen bundle architecture, along with reduced or absent elastic fibers. Stains for mucin and lipid are frequently positive in these areas.

 **FIGURE 43-4** Palisading granulomatous inflammation surrounding degenerating collagen within the dermis. (Hematoxylin and eosin stain, $\times 200$.) (Used with permission from Dr. Richard Crawford.)

Immunofluorescence and Ultrastructural Findings

Direct immunofluorescence may reveal deposition of fibrin, immunoglobulin M (IgM), and complement component C3 around dermal blood vessel walls or at the basement membrane zone, although these findings are variable. IgM-positive cytol bodies have also been reported. Ultrastructural studies have demonstrated abnormalities in connective tissue and capillaries.

Special Tests

Diagnosis of granuloma annulare is typically established clinically or confirmed by skin biopsy. Routine special investigations are generally not required. However, in atypical presentations, further evaluation may be warranted to exclude systemic conditions such as infection, sarcoidosis, or malignancy. Assessment for endocrine disorders—particularly diabetes mellitus or thyroid dysfunction—should be considered if clinical signs or symptoms are present.

In cases of subcutaneous granuloma annulare with atypical features (e.g., rapid growth, pain, or unclear clinical diagnosis), imaging studies may be useful:

- **Radiographs:** Show a non-specific soft tissue mass without calcification or bone involvement.
- **Ultrasonography:** Reveals a hypoechoic lesion in the subcutaneous fat.
- **MRI:** Demonstrates a mass with indistinct margins, isointense or slightly hyperintense to muscle on T1-weighted images, and heterogeneous with predominantly high signal intensity on T2-weighted images.

Differential Diagnosis

Refer to Box 43-1 for differential diagnoses by clinical variant.

Box 43-1: Differential Diagnosis of Granuloma Annulare

Annular Type

Consider: - Tinea corporis

- Subacute cutaneous lupus erythematosus
- Neonatal lupus erythematosus
- Annular lichen planus
- Acute febrile neutrophilic dermatosis (Sweet syndrome)
- Erythema chronicum migrans
- Actinic granuloma
- Necrobiosis lipoidica diabetorum

Rule Out: - Infections (e.g., tuberculosis, atypical mycobacteria, syphilis)

- Interstitial granulomatous dermatitis with arthritis
- Interstitial granulomatous drug reaction
- Annular sarcoidosis
- Lymphoma

Generalized Type

Consider: - Lichen planus

- Lichen nitidus
- Molluscum contagiosum

Rule Out: - Lichenoid and granulomatous dermatitis associated with HIV/AIDS

- Infections (e.g., tuberculosis, atypical mycobacteria, syphilis)
- Sarcoidosis
- Blau syndrome (familial granulomatous arthritis, skin rash, uveitis)
- Interstitial granulomatous drug reaction
- Lymphoma

Subcutaneous Type

Consider: - Erythema nodosum

- Dermoid cyst
- Rheumatoid nodules

Rule Out: - Epithelioid sarcoma

- Benign or malignant soft tissue tumors
- Deep fungal or bacterial infections

Perforating Type

Consider: - Molluscum contagiosum

- Insect bites
- Pityriasis lichenoides
- Perforating collagenosis and other perforating disorders
- Foreign body granuloma
- Papulonecrotic tuberculid
- Palisaded neutrophilic and granulomatous dermatitis (associated with immune complex diseases)

Patch Type

Consider: - Morphea (localized scleroderma)

- Erythema annulare centrifugum
- Parapsoriasis

Rule Out: - Lymphoma

Treatment

Refer to Box 43-2 for treatment options.